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Active tokens and crypto-asset valuation

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Abstract

This research investigates token dormancy as a fundamental metric for evaluating cryptocurrency assets and presents a methodology for its measurement. The valuation method involves 4 distinct parameters and utilizes a 3.5-year daily dataset for the “Chainlink” token. The results are used in optimized ARIMA-GARCH models to analyze the first differences between the variables; the out-of-sample forecasts were assessed with performance metrics. Furthermore, this study introduces a novel fundamental value derived from these approaches, the basis for generating selling signals in a backtested trading strategy. The trading strategy results are compared to a benchmark buy-and-hold strategy and a non-selling dollar-cost-averaging strategy for evaluation. Employing the dollar-cost averaging approach for purchase frequency and utilizing the “isolation forest” technique for identifying selling signals within the trading strategy yielded positive results.

Keywords: Crypto-asset valuation, Crypto-asset activity, Crypto-asset dormancy, ARIMA-GARCH, Backtesting trading strategy, Fundamental value

Introduction

A fundamental factor in analyzing a crypto-asset is the valuation method used to determine its approximate fair price; suggested valuation approaches include a market, a cost, and an income approach (Johnson et al. 2019). However, crypto-assets do not exhibit the same attributes as traditional assets; thus, their valuation is controversial (Drofenik and Kovačič Batista 2018). For example, under certain conditions, some crypto-assets provide the holder with income when their protocol uses “proof of stake” (PoS) consensus protocols, like Ethereum (Kapengut and Mizrach 2023). Furthermore, Bitcoin has a production cost that could be reflected in its price because of its “proof of work” (PoW) consensus protocol, which requires specialized hardware for mining new Bitcoin and validating the network’s transactions (Kristoufek 2015). Therefore, monitoring the marginal production costs of a crypto-asset in real time would be helpful when fundamentally evaluating Bitcoin (Hayes 2019). However, methods of evaluating tokens that do not provide an income (such as in PoS staking) or require a constant flow of electricity and computing power for production (as in the PoW protocol) remain debated.

The most popular methods for valuing a crypto-asset include market capitalization (MC) and fully diluted valuation (FDV) methods. Calculating FDV and MC may seem straightforward; however, specific prerequisites are involved in obtaining the required

variables for their computation. Regarding the market price, unlike an organized stock market that provides the current market price or the closing price, no such market exists for a crypto-asset (CA) because multiple exchanges operate worldwide on a 24-h basis. Some exchanges even produce fake trading volumes (Chen et al. 2022), creating doubts about data reliability. Instead, the price reported from one or multiple exchanges of choice could be used, or a data-aggregator service, such as Coingecko.com or Coinmarketcap.com, could be utilized, and their respective methodology could be followed when calculating the CA's price.

Regarding the circulating or total supply of the CA, one may still rely on data-aggregator services to be informed about these values. Because the CA market is unregulated, we must have accurate and valid information about the CA supply to evaluate because the token supply might be conditionally dynamic or static. Private keys for crypto wallets can be lost in numerous ways (Hinkes 2019); for example, 20% of Bitcoin in circulation did not move for five years and was considered lost (Chainalysis 2020). Similar cases are also possible in other blockchain networks, when tokens may remain dormant indefinitely. In the Ethereum network, there may be cases where tokens are sent to smart contracts or addresses where retrieving such tokens is impossible because of programming limitations and errors (Piantadosi et al. 2023), smart contract exploits (C. Liu et al. 2018), or delivery to incorrect addresses whose private keys are not known. Therefore, wallet dormancy may be used to predict whether access to a wallet is lost (Pokharel and Dahal 2022). Additionally, using dormancy derivative metrics, such as the Bitcoin Realized HODL Ratio, helps to identify the Bitcoin cycle tops (Swift 2020). Assuming some Bitcoins were known to be lost, should they be included in the valuation of the Bitcoin network using the market capitalization approach?

The concept of network effects suggests that the value of a product or service increases as its user base grows. Metcalfe's law (Shapiro and Varian 1999) represents the impact of network effects, asserting that a network's value is proportional to the square number of its participating users. Metcalfe's law has been studied for valuing Bitcoin by authors such as Van Vliet (2018) and Alabi (2017); hence, it may be beneficial to devise a methodology for estimating the active users of a network or employ a measure that can serve as a proxy for such activity, which could then be utilized in a valuation approach for crypto-assets.

By understanding the importance of dormancy calculation in crypto-assets, this study was motivated to explore a method that examined the dormancy of every circulating token and represented the genuine market value of a crypto-asset based on its token activity. Therefore, this study investigates how token dormancy can be incorporated into the valuation of crypto-assets and whether it holds any significant value or utility.

This research paper proposes a method that calculates an activity score for every address that holds the examined token. This study applies a new method, called "active token balance" (ATB), to demonstrate the new market capitalization method called "active market capitalization" (AMC). Finally, this study proposes a novel fundamental value method called "active token price" (ATP). The methodology is demonstrated using actual data from Blockchair.com, covering 3.5 years for the token "Chainlink" (LINK), which is deployed on the Ethereum network (Appendix 3). Subsequently, the autoregressive integrated moving average-generalized autoregressive conditional

heteroskedasticity (ARIMA-GARCH) models are applied with the ATB series as exogenous variables for estimation and forecasting. The ATP values are then utilized in a backtesting trading strategy, and their results are evaluated.

To conclude, this research contributes to the field by introducing a customizable tool that improves the understanding of a crypto-asset's overall behavior, which can be tailored to individual preferences and applied in forecasting. Furthermore, this paper refines the market valuation of the crypto-asset by considering potentially lost tokens, offering a valuable metric for trading and evaluation.

The manuscript begins with a comprehensive literature review focusing on the significance of crypto-asset activity and dormancy. The methodology section provides a succinct overview of the sample selection and cleaning process, the dormancy measurement technique, and the valuation methodology. This section also outlines the methodology governing the estimation of the ARIMA-GARCH models and the setup of the backtesting trading strategy. The results section sequentially presents the outcomes of the valuation methodology, followed by the findings derived from the ARIMA-GARCH estimations. The results section concludes with the outcomes of the backtesting strategy. Subsequently, the discussion section explores the pivotal facets of these results, and the conclusion section delineates the implications for the respective field and expounds upon the findings. Finally, three appendix sections contain additional formulas, figures, and supplementary explanatory notes.

Literature review

Determining an asset's value is an ongoing question among financial analysts, researchers, and investors. While some widely used methods are available to assess the value of various assets, the relatively new crypto-asset industry might need traditional valuation methods or the creation of new ones. Regarding the cost of production theory, Hayes (2019) found that Bitcoin's market price fluctuated around the predicted model price, incorporating a marginal production cost; however, Hayes recognized that his method could not necessarily explain the prices of Ethereum and Litecoin. The findings in the previously mentioned research are essential for the case of Bitcoin; however, the cost of the production model cannot be effectively applied in cases where the token creation or the consensus protocol uses minimal electricity, as in the case of a PoS consensus protocol.

Bhambhwani et al. (2021) supported the idea that the network strength and a blockchain's computing power show intrinsic value in a crypto-asset; however, they acknowledged that their research did not include the non-mineable tokens' hash rate because such a metric was non-existent and could not be used for this category of CAs. Cong et al. (2021) developed a model that considered the transactional activity of different users and explained the dynamics of a token value influenced by the users' participation. The transactional activity of Bitcoin could be a helpful indicator when trying to evaluate its potential returns (Koutmos 2018); thus, monitoring such activity could provide helpful information about its fundamentals. The "coin days destroyed" (CDD) indicator is a measure of Bitcoin activity; CDD equals the number of days from the date on which the unspent transaction output (UTXO) was created, multiplied by the amount of Bitcoin in the UTXO (Bytecoin 2011). CDD focuses

on the dates when UTXOs are spent, not on the wallet balances or the rest of the UTXOs' dormancy in a wallet. Furthermore, this metric does not connect the total dormancy of coins in the Bitcoin network; it is a metric of the assumed strength of the spent outputs and the change in Bitcoin holder behavior.

Hauge (2019) introduced a derivative metric of CDD called supply-adjusted coin days destroyed, which accounts for how the CDD metric impacts the circulating supply of Bitcoin; however, this metric provides no insight into the dormant state of the remaining Bitcoin. Bansal (2018) developed the "HODL waves" model, which presents the age distribution of the UTXOs in the Bitcoin network and aggregates them into different age bands. In the "HODL waves" model, whenever a UTXO is used in a transaction, it is considered active again and falls into the most active age band, resetting its state of dormancy. This method is useful for observing the dormancy of UTXOs; however, it is possible to spend only specific UTXOs from a Bitcoin wallet. Therefore, this method does not show whether the remaining UTXOs belong to an active user and does not connect Bitcoin value with dormancy. Shirakashi (2019), as cited in Glassnode (2022), created the average spent output lifespan (ASOL), which measures the average age, in days, of all the Bitcoins spent in a transaction period. According to Glassnode, when the ASOL value reaches 40 days, it correlates with local market tops or bottoms, and high volatility is observed for values of over 60 days. Smith (2018) computed the "Bitcoin average dormancy" metric by examining the ratio of the total CDD to the Bitcoin transaction volume. Their findings indicated that dormant coins, unused for an extended period, were reintroduced into circulation at elevated prices.

Due to challenges in accurately identifying active participants because of privacy concerns and the potential for users to create multiple wallets, Alabi (2017) used the number of unique addresses as a proxy for the number of active users. Furthermore, in their research on network effects in cryptocurrencies, Stylianou et al. (2021) asserted that when employing a surrogate indicator for a cryptocurrency's user base, they considered dormant and permanently inaccessible addresses economically insignificant.

Liebi (2022) proposed a valuation metric known as network value, which is the ratio between the 30-day moving average active addresses to the market capitalization of the examined crypto-asset. They found that crypto-assets with high respective ratio values have higher average returns than those with low values.

Furthermore, Bakhtiar et al. (2023) utilized active wallet addresses as a substitute for the node component in the context of Metcalfe's law. They also observed a positive impact of the number of active addresses on most cryptocurrency prices they investigated, arguing for incorporating inactive and staked users into a novel network valuation model. They asserted that this approach would offer a more precise depiction of the actual value of cryptocurrencies, and a factor with the potential to predict cryptocurrency prices is the concept of "active supply."

This study's literature review revealed a research gap regarding a metric to measure the dormancy state of the tokens in every address and how this state could be reflected in a valuation method.

Methodology

Collect and prepare the data

The data were collected from Blockchair.com, including all ERC-20 daily transactions from the Ethereum blockchain from September 16, 2017, to February 3, 2021. The data were filtered to isolate 7,051,242 transactions of the LINK smart contract (Appendix 3).

This study used the Pandas module in Python (McKinney 2010; The Pandas Development Team 2020) to organize these transactions according to their respective dates, totaling 1237 days. Every row included the (1) transaction date and time, (2) sender address, (3) recipient address, and (4) token quantity. Afterward, 1237 tables were created to present the daily transaction ledger for every involved address on the examined dates and their respective (1) deposits, (2) withdrawals, and (3) daily final balances. The daily transaction tables were used to create, update, and save 1237 daily tables that reported the final daily balance of every address with a positive token balance on the current or the previous date. An additional variable showed if the balance of the address, for the corresponding table date, had either (1) only increased, (2) decreased (even if it had increased in the exact date), or (3) remained unchanged. Finally, the token LINK's market capitalization, price, and total supply data were obtained from Coingecko.com (2022).

Calculating dormancy

This study used the data reported from the final daily balance tables to calculate the dormancy of the tokens in every i address for each corresponding t date. If a wallet holder has sent tokens, then the address owner is assumed to (a) be active and still interested in the token, (b) hold the private keys of the sender address, which means that the tokens are accessible, and (c) use the tokens because they are economically important enough to motivate their owner to spend “gas” (i.e., transaction costs on the Ethereum blockchain) in Ether to execute the transaction. Therefore, the tokens in the sender's address are considered active and have zero dormancy. The tokens in the remaining addresses that have not moved in an examined t date are penalized to their dormancy score. The “transaction status” (TrS) variable in the final daily balance table receives four integer values, from “0” to “3,” as shown in Formula (1):

$$Dormancy_{i,t} = \begin{cases} D(0)_{i,t} \Rightarrow \text{if the address emerged for the first time} \\ D(1)_{i,t} \Rightarrow \text{if the address had a positive balance of tokens and no transactions} \\ D(2)_{i,t} \Rightarrow \text{if the address had sent tokens} \\ D(3)_{i,t} \Rightarrow \text{if the address had a deposit of tokens and no withdrawals} \end{cases} \quad (1)$$

The integer values from “0” to “3” are qualitative variables for the four cases when calculating the final dormancy of the tokens in an address; each case corresponds to a TrS value.

Case 1: First-time emerging address

If an address is presented in the daily tables for the first time, its tokens' dormancy is set to 0, and the TrS has a value of “0,” as in Formula (2).

$$D(0)_{i,t} = 0 \quad (2)$$

Case 2: The address had a positive balance of tokens and no transactions

The tokens' dormancy is increased by 1, representing 1 elapsed period, regarding the dormancy of the previous date with a TrS of "1," as in Formula (3).

$$D(1)_{i,t} = Dormancy_{i,t-1} + 1 \quad (3)$$

Case 3: The address has sent tokens

An address with at least one withdrawal transaction is assigned a TrS value of two, and the dormancy of its tokens is set to zero, as in Formula (4).

$$D(2)_{i,t} = 0 \quad (4)$$

Case 4: The address had a deposit of tokens and no withdrawals

With a TrS of "3," the dormancy of the newly deposited tokens is 0, and the dormancy of the old tokens is increased by 1 elapsed period. The address's new total dormancy is a weighted average of the old and the new tokens' dormancy, which considers the volume of new tokens and incorporates them into the calculation. The deposited tokens (DT) in time t in the address have zero dormancy. The equations for this case are shown in Formula (5):

$$\begin{aligned} D(3)_{i,t} &= \frac{[(Dormancy_{i,t-1} + 1) \cdot Balance_{i,t-1}] + (0 \cdot DT_{i,t})}{Balance_{i,t-1} + DT_{i,t}} \Leftrightarrow \\ &\Leftrightarrow D(3)_{i,t} = \frac{(Dormancy_{i,t-1} + 1) \cdot Balance_{i,t-1}}{Balance_{i,t}}. \end{aligned} \quad (5)$$

The final dormancy value for an address can range from zero upwards. This metric indicates the actual duration in the measured time frequency for which all the tokens remain inactive in an address.

Calculating the dormancy penalty

Next, the dormancy penalty P for every i address in the t date is calculated as in Formula (6).

$$P_{i,t,LL} = \min\left(\frac{Dormancy_{i,t}}{Useful\ life_{CA}}, 1\right), P \in [0, 1] \quad (6)$$

The final dormancy value of an address is utilized in the numerator of the "dormancy to useful life" ratio, which may be equal to zero or even exceed one, as the dormancy value can be zero or surpass the useful life value, respectively. Therefore, the penalty ranges from 0 to 1, with a minimum value of 1 established in Formula (6), meaning that if the dormancy value exceeds the "useful life" (UL) value, the penalty is no greater than 1. This study uses different values of UL: 30, 60, 120, and 180 days for the token. Furthermore, Appendix 1 shows that a unity-based normalization approach can be used to calculate the penalty value.

The address's dormancy penalty, denoted as P , is subtracted from 1, the value that equals zero dormancy. Then, each wallet's final activity score (AS) is calculated for each date, as shown in Formula (7).

$$AS_{i,t,UL} = 1 - P_{i,t,UL}, AS \in [0, 1] \quad (7)$$

Therefore, the maximum activity score equals one; similarly, the maximum penalty score equals one because the tokens in a wallet might be dormant for more than the assumed UL. For example, a dormancy of more than 180 days with a UL setting of 180 would result in a penalty value larger than one, leading to a negative score that would negatively affect the rest of the network's total activity.

Calculating the activity-scored address balance

After obtaining the scores for every address, this study calculated the activity-scored address balance (AB) by multiplying the address activity score and the address final token balance with LINK for every address i on a date t in Formula (8).

$$AB_{i,t,UL} = AS_{i,t,UL} \cdot Balance_{i,t} \quad (8)$$

Subsequently, Formula (9) calculates the total ATB for the token LINK for every i address on a date t :

$$ATB_{LINK,t,UL} = \sum_{i=1}^n AB_{i,t,UL}. \quad (9)$$

Moreover, the final dormancy-based market capitalization based on the activity of the token LINK (AMC) is the product of the current market price (MP) times the active token balance of LINK as in Formula (10):

$$AMC_{LINK,t,UL} = ATB_{LINK,t,UL} \cdot MP_{LINK,t} \quad (10)$$

Token fundamental value

The proposed fundamental value ATP is the ratio between the token's ATB and the circulating supply (CS), defined as the "ATB Ratio," multiplied by the current market price, as in Formula (11).

$$ATP_{LINK,t,UL} = ATB_{Ratio_{LINK,t,UL}} \cdot MP_{LINK,t} = \left(\frac{ATB_{LINK,t,UL}}{CS_{LINK,t}} \right) \cdot MP_{LINK,t} \quad (11)$$

Token capitalization methods

This study uses two main methods, market capitalization (MC) and fully diluted valuation (FDV), to assess the relative size and valuation of crypto-assets within the cryptocurrency market. MC and FDV are presented in Formulas (12) and (13), respectively.

$$MC = Market\ Price \cdot Circulating\ Supply \quad (12)$$

$$FDV = Market\ Price \cdot Total\ Supply \quad (13)$$

MC is the product of the market price times the CS of the token, and FDV is the market price times the token's total supply. By rearranging the market capitalization Formula (12), we obtain the CS at time t as follows:

$$\text{Circulating Supply}_t = \frac{MC_t}{\text{Market Price}_t} \quad (14)$$

This study calculated the FDV using Formula (13) and the daily CS of the token using Formula (14).

Analysis using ARIMA and GARCH

This study employed the augmented Dickey–Fuller (ADF) test from the “statsmodels” module in Python to examine the presence of a unit root. The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test was also used to assess stationarity in the time series (Seabold and Perktold 2010). If stationarity is not present, then the time series variable “X” under examination (which may be the market price or variants of ATB) is transformed into Formula (15) of the logarithmic differences:

$$\text{First difference in } \log(X_t) = \ln(X_t) - \ln(X_{t-1}). \quad (15)$$

Data that became stationary after the first difference were fitted using an automated ARIMA procedure employing R’s “rugarch” package to ensure consistency among the tested observations (Engle and Sokalska 2012; Ghalanos 2022). Additionally, exogenous variables were incorporated into the following general model.

$$r_{\text{Pr}_{\text{icet}}} = \mu + \sum_{i=1}^p \varphi_i r_{\text{Pr}_{\text{icet}}-i} + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \sum_{i=1}^a \beta_i x_{t-i} + \varepsilon_t \quad (16)$$

In Formula (16), r_{Price} is the return of the price of Chainlink at time t , μ is the constant, and φ is the coefficient of the AR term. θ is the coefficient of the MA term, and β is the coefficient of the optional exogenous variable. The maximum likelihood estimation method was utilized to estimate the parameters of the ARIMA model. The residuals were assumed to follow a t-student distribution with a mean of zero, σ^2 variance, and ν degrees of freedom. The process evaluates the optimal model by testing various combinations of up to two autoregressive (AR) terms, two moving average (MA) terms, and up to 3 lags for the variable under examination (Fig. 1).

Subsequently, the top-performing models were examined for autoregressive conditional heteroskedasticity (ARCH) effects. ARCH effects were detected; therefore, this study used a GARCH model with t-student distribution fitted standardized residuals, known as GARCH-t (Bollerslev 1987), utilizing the corresponding best ARIMA

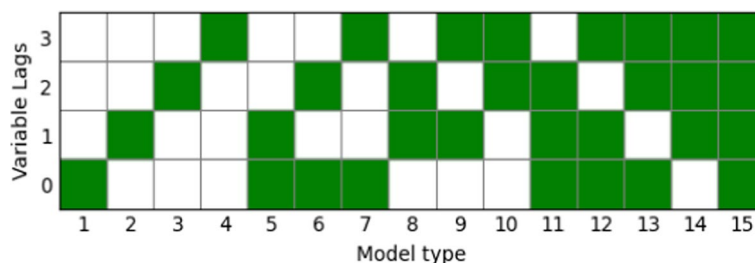


Fig. 1 Lags of exogenous variables across different models

model with exogenous variables as the mean equations. The general form of the GARCH(1, 1) equation under examination i is specified in Formula (17).

$$\begin{aligned} r_{Prict} &= \mu^* + \varepsilon_t \\ \varepsilon_t &= e_t \sigma_t \\ \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned} \quad (17)$$

The conditional mean of the model contains μ^* , which is the optimal model picked by the automated ARIMA procedure described in (16). σ_t^2 is the conditional variance at time t , ω is the intercept, and α is the coefficient for the lagged conditional error term ε_t^2 , which represents the impact of past volatility shocks. β is the coefficient for the lagged conditional variance σ_t^2 , which captures the persistence of past volatility (e_t) a sequence of variables assumed to follow a t-student distribution with a mean of zero, σ^2 variance, and ν degrees of freedom.

Forecasting and performance metrics

This study used ARIMA-GARCH models to encompass out-of-sample forecasts for the mean of LINK returns, extending until February 18, 2021, spanning up to 15 days. The accuracy of the results was assessed through the mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and mean absolute percentage error (MAPE) metrics (Appendix 1).

Finally, this study employed the MCS test to evaluate the forecasting performance of the tested models (Hansen et al. 2011). The MCS procedure involves selecting the superior set of models M^* (SSM) from the initial set (M), for which the null hypothesis of equal predictive ability (EPA) is not rejected for a specified significance level of α . No specific consensus exists regarding the optimal value of the α parameter; hence, this study set this parameter to 10%, following Maciel et al. (2022). The absolute error (AE) and squared error (SE) loss functions are employed in the MCS method. The test statistic used for the test is T_{max} , with a bootstrap value of 5000 resampling for constructing the statistical test, following the methodology as reported in the “MCS” package in R language (Bernardi and Catania 2018).

Backtesting strategy using the new fundamental prices

This section delineates the backtested trading strategies: “buy-and-hold,” dollar-cost averaging without selling, and incorporating the difference between the LINK price and ATP in the sell signals.

Assume three separate entities invest in LINK: “BH,” which buys and holds, and “A” and “B,” which use the dollar-cost averaging (DCA) method when buying the token, which is backtested daily up to a 30-day basis for fixed amounts of 10 US dollars (USD). The DCA method is a trading strategy where an entity may invest equal amounts of money in an asset over regular intervals. For this comparison of backtesting strategies, only these intervals are considered to be “buy” signals for “A” and “B.”

The examined entities (“A” and “B”) start with an initial investment of 10 USD and employ the DCA strategy for buying the asset from the next day forward. Entity “BH” follows a never-selling buy-and-hold strategy. Furthermore, entity “BH” invests the same

amount of money on the initial day of the testing period that the other entities invest in the entire period. Entity “A” uses the non-selling DCA strategy, and “B” buys at the same time that “A” does but sells whenever it receives a “sell” signal. Whenever the next “buy” interval is due, “A” rebuys LINK worth 10 USD, and “B” buys LINK worth 10 USD plus any funds accumulated from the previous sale. Entity “B” bases its sell strategy on monitoring the difference between the LINK market price and ATP; it sells whenever an “event” occurs.

Event definition and detection

An “event” is an observed outlier, in time t , among a list containing the differences between the market price and ATP. The event occurs if the outlier has a negative value, which denotes a sudden and unexpected increase in ATP.

The first step to finding an “event” is calculating the difference between the ATP and the MP, as seen in Formula (18).

$$d_{(MP_{LINK,t}-ATP_{t,ul}),t} = MP_{LINK,t} - ATP_{t,ul} \quad (18)$$

Next, this study calculates the difference of Formula (18) to explore the variation of this difference over time:

$$dd_{(MP_{LINK,t}-ATP_{t,ul}),t} = d_{(MP_{LINK,t}-ATP_{t,ul}),t} - d_{(MP_{LINK,t}-ATP_{t,ul}),t-1} \quad (19)$$

The results of Formula (19) show how much the difference of Formula (18) has changed over time, regardless of whether the “gap” between the fundamental price and the market price has increased.

This study uses the “Isolation Forest” method (Liu et al. 2008) to determine the outlier values, as implemented in the Scikit-Learn module in Python (Pedregosa et al. 2011). The “isolation forest” method relies on the concept that anomalies are less frequent and have attributes that differ from normal data points. The fitting of this technique returns anomaly scores of 1 for inliers and -1 for outliers for every value in the dataset. In this study, a “contamination” level of 5% is chosen as a function argument, the percentage of outliers present in the dataset under examination. In the “isolation forest” function, the random state argument is set to “123” for reproducible results; every other argument is set to default. The “isolation forest” technique is used in the values of a rolling window, which is a one-dimensional list that contains n observations of Formula (19) values that range from n values of 3 to 30 regarding the test conducted. This test is as follows:

$$window_{dd_{(MP_{LINK,t}-ATP_{t,ul}),t}}^{k=1} = \left\{ dd_{(MP_{LINK,t}-ATP_{t,ul}),t-(n-k)}, \dots, dd_{(MP_{LINK,t}-ATP_{t,ul}),t} \right\}. \quad (20)$$

Formula (20) depicts the general window formula, where k is the position of the observation in the list, and t is the time of the observation. In this backtesting strategy, for each day analyzed, this method utilizes the window of n observations, and “B” verifies whether the observation is an outlier at time t . If the observation in time t is an outlier with a negative value, it is an “event” and prompts a “sell” signal.

The first date of the backtesting is November 10, 2017, the first date where an observation “window” regarding Formula (20) exists; the first token purchase occurs on the same day for both entities “A” and “B.” February 3, 2021, is the last date in this study’s calculations regarding ATP.

Backtesting results and metrics

The final results regarding the different trading strategies are reported according to the total investment portfolio returns and their respective Sharpe ratios (Sharpe 1966). The Sharpe ratio is defined in Formula (21) as follows:

$$Sharpe_j = \frac{R_j}{\sigma_j}. \quad (21)$$

Here, j is the portfolio, R is the portfolio’s returns after accounting for the 3-month US T-bill rate during the testing period, and σ is the standard deviation of the portfolio’s returns. The Sharpe ratio is calculated using daily logarithmic returns and is reported without annualization.

Results

This section initially presents the results for AMC, ATB, and ATP concerning LINK. This study then provides statistical insights into the selected variables earmarked for further analysis using the ARIMA model. This approach lays the foundation for implementing the GARCH model and uses the estimations for forecasting and performance evaluation. Subsequently, the backtesting results of the trading strategy are revealed, encompassing the total investment and benchmark outcomes, event count, test strategy performance, and the corresponding Sharpe ratios for the examined entities.

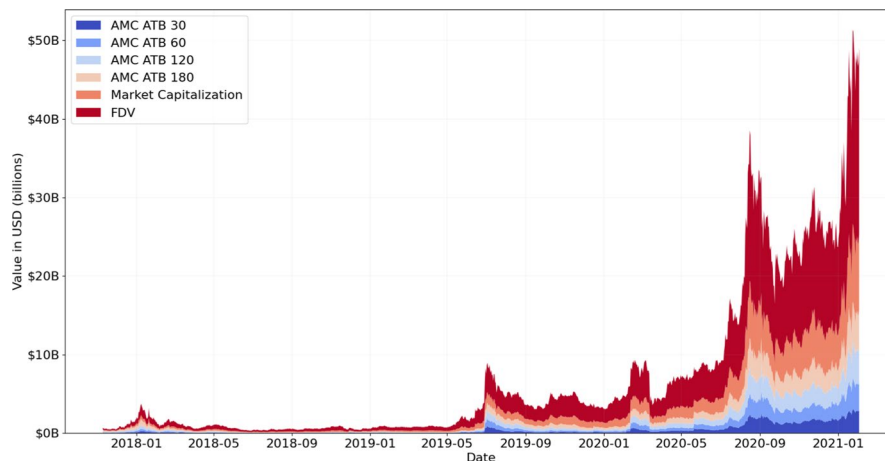


Fig. 2 Market Capitalization Methods Comparison

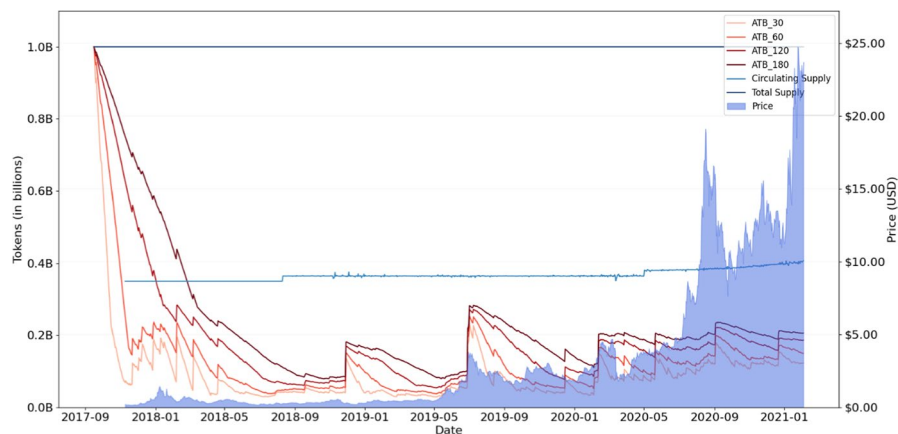


Fig. 3 Token supply and price

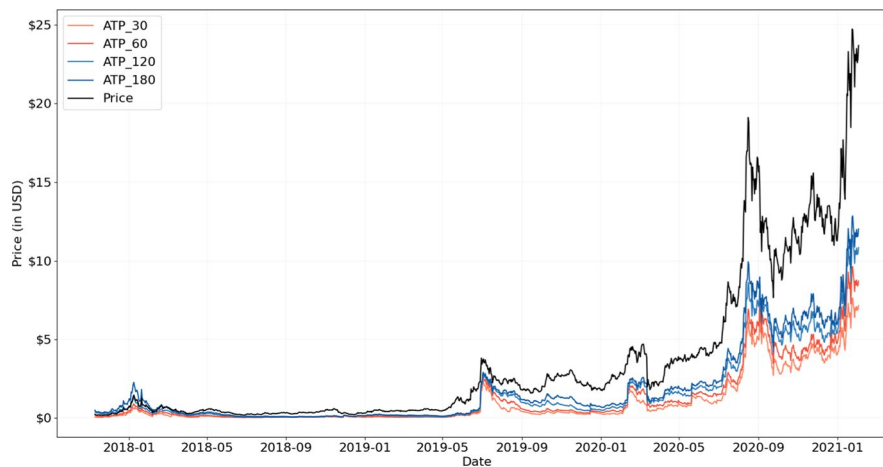


Fig. 4 ATP Variants and price

The results for the AMC of LINK

The results derived from Formulas (12), (13), and (10) were plotted using the “Matplotlib” module in Python (Hunter 2007). Figure 2 presents these results.

The results show that the AMC method gives lower capitalization values than MC and FDV because the quantity of the tokens assumed to be active is lower than the other two methods. Decomposing AMC into ATB and the price of LINK clearly illustrates the visual distinctions of the token supply metrics (Fig. 3).

ATB results

Figure 3 shows that the total supply of LINK is steady at 1 billion tokens. The CS was calculated using Formula (14), per the data from Coingecko; the CS started at 350 million tokens and ended at approximately 0.4 billion tokens, with a slightly positive trend. ATB implements Formula (9) and represents the total number of active tokens based on this study’s method. Initially, the ATB balances (representing the active tokens) rapidly declined until they stabilized from approximately 50 million to 200 million tokens. Afterward, the token’s activity spiked around December 2018, July 2019, February 2020,

Table 1 ADF and KPSS Stationarity tests

Variable	ADF		KPSS	
	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
ATB_30	− 3.52	0.01***	1.09	0.01***
ATB_60	− 2.86	0.05**	0.71	0.01***
ATB_120	− 5.29	0.00***	0.75	0.01***
ATB_180	− 5.22	0.00***	1.16	0.01***
r_ATB_30	− 33.6	0.00***	0.03	0.10
r_ATB_60	− 32.96	0.00***	0.17	0.10
r_ATB_120	− 17.69	0.00***	0.38	0.08*
r_ATB_180	− 17.31	0.00***	0.41	0.07*
Price	1.16	1.00	3.76	0.01***
r_Price	− 18.88	0.00***	0.08	0.10

Asterisks indicate significance levels: * indicates $p < 0.10$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$. The ADF test null hypothesis is that the time series has a unit root. The KPSS test null hypothesis is that the times series is stationary

Table 2 Descriptive statistics of the test variables

Variable	Observations	Mean	SD	Max	Min	Skewness	Kurtosis	Jarque–Bera	Jarque–Bera <i>p</i> -value
r_ATB_30	1182	0.00	0.09	1.22	− 0.12	8.23	86.85	359,650.88	0.00***
r_ATB_60	1182	0.00	0.06	1.01	− 0.09	10.60	144.45	1,007,475.00	0.00***
r_ATB_120	1182	0.00	0.04	0.84	− 0.05	14.44	248.26	3,003,636.00	0.00***
r_ATB_180	1182	0.00	0.04	0.76	− 0.04	16.93	323.90	5,128,144.00	0.00***
r_Price	1182	0.00	0.08	0.48	− 0.66	0.12	11.49	3555.87	0.00***

Asterisks (***) denote significance at the 1% level

September 2020, and December 2020, and significant volatility clustering in the price of LINK existed around these periods.

ATP results

Figure 4 displays the outcomes derived from Formula (11), utilizing the ATB values shown in Fig. 3 in conjunction with LINK's market price. The plot reveals that LINK's market price remains consistently higher than the other ATP variants, indicating that LINK traded at a premium for the examined period. Conversely, during the initial period, ATP 180 exhibited higher prices, suggesting that during that time, the LINK token traded at a discount based on these values.

Variable selection

Table 1 presents the results of the stationarity tests conducted on the ATB variant variables, which are used in the subsequent ARIMA-GARCH analysis.

The dependent variable (*r_Price*) represents LINK's market price returns. The independent variables selected for the models include *r_ATB_30*, *r_ATB_60*, *r_ATB_120*, and *r_ATB_180*. The selected variables correspond to the first differences in the ATB time

series based on their UL parameters. Table 2 presents the descriptive statistics for variables analyzed in the ARIMA-GARCH models.

The mean values were close to zero across all variables, indicating a relatively balanced distribution around the center. The standard deviations are relatively small, suggesting limited variability. Skewness and kurtosis values indicate departures from a normal distribution, with positive skewness and high kurtosis values indicating heavy tails and peakedness. The Jarque–Bera (JB) test results reject the null hypothesis of normality for all variables, as evidenced by extremely low p-values. The initial dataset contained 1236 observations for the ATB differences values and 1182 observations for the price returns values. The tested data was reduced to 1182 observations for all variables to ensure consistency, covering November 10, 2017, to February 3, 2021. The selected variables are used in the subsequent ARIMA-GARCH models.

ARIMA results

Using the Akaike information criterion (AIC), this study selected the following ARIMA models that contained the relevant exogenous variable under examination.

Table 3 shows that “7” is the optimal ARIMA model incorporating the exogenous variables r_ATB_30 and r_ATB_60 . For the variables r_ATB_120 and r_ATB_180 , the best ARIMA model is “6.” Regarding the best ARIMA model for variable r_Price alone, the best model free of diagnostic errors during the ARIMA and GARCH estimation follows an ARIMA (0, 0, 2) process with an AIC value of -2.5125 . Table 4 presents the results of the best ARIMA models, demonstrating the importance of the exogenous variables in most cases and the significant AR(1) terms in models two, three, four, and five. Model 1 features a significant MA(1) term instead; however, all models exhibit ARCH effects.

Table 3 AIC values of the best ARIMA models

Model	Exogenous variable			
	r_ATB_30	r_ATB_60	r_ATB_120	r_ATB_180
1	− 2.5262	− 2.5267	− 2.5273	− 2.5283
2	− 2.5158	− 2.516	− 2.517	− 2.5177
3	− 2.516	− 2.5154	− 2.5168	− 2.5177
4	− 2.518	− 2.5178	− 2.5171	− 2.5167
5	− 2.53	− 2.5297	− 2.5309	− 2.5321
6	− 2.5299	− 2.5291	− 2.5317*	− 2.5335*
7	− 2.5313*	− 2.5302*	− 2.5306	− 2.5311
8	− 2.5144	− 2.5139	− 2.5165	− 2.5182
9	− 2.5163	− 2.5162	− 2.5165	− 2.5168
10	− 2.5168	− 2.5162	− 2.5171	− 2.5174
11	− 2.5285	− 2.5274	− 2.5305	− 2.5328
12	− 2.5299	− 2.5285	− 2.5292	− 2.5301
13	− 2.53	− 2.5286	− 2.5307	− 2.532
14	− 2.5151	− 2.5147	− 2.5169	− 2.518
15	− 2.5287	− 2.5269	− 2.5295	− 2.5313

The asterisk (*) indicates the model with the lowest AIC among those considering the exogenous variable. The model numbers correspond to the tested models, as presented in Fig. 1

Table 4 ARIMA Analysis: r_Price with ATB Variants

Model number / Exogenous variable	mu	ar (1)		ma (1)		ma (2)		Var_0		Var_1		Var_2		Var_3	
		Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value
1. (r_Price)	0.001 (0.002)	0.468 [0.640]		− 0.087 (0.039)	− 2.211 [0.027]**	0.046 (0.032)	1.451 [0.147]								
2. r_ATB_30	0.002 (0.002)	0.929 [0.353]	− 0.088 (0.039)	− 2.253 [0.024]**				0.100 (0.030)	3.345 [0.001]***					0.028 (0.019)	1.510 [0.131]
3. r_ATB_60	0.002 (0.002)	0.946 [0.344]	− 0.086 (0.039)	− 2.197 [0.028]**				0.143 (0.055)	2.614 [0.009]***					0.036 (0.031)	1.133 [0.257]
4. r_ATB_120	0.002 (0.002)	0.936 [0.349]	− 0.090 (0.040)	− 2.263 [0.024]**				0.231 (0.120)	1.919 [0.055]*			− 0.052 (0.034)	− 1.528 [0.126]		
5. r_ATB_180	0.002 (0.002)	1.001 [0.317]	− 0.091 (0.040)	− 2.300 [0.021]**				0.296 (0.100)	2.946 [0.003]***			− 0.073 (0.036)	− 2.049 [0.041]**		
Model number / Exogenous variable	Sigma	Shape		ARIMA order		ARCH LM tests									
		Estimate	t value	Estimate	t value	Lag 2	Lag 5	Lag 10							
1. (r_Price)	0.079 (0.006)	14.070 [0.000]***	3.380 [0.359]	9.410 [0.000]***	(0, 0, 2)	9.443 [0.009]***	15.908 [0.007]***	19.982 [0.029]**							
2. r_ATB_30	0.078 (0.006)	14.084 [0.000]***	3.402 (0.361)	9.428 [0.000]***	(1, 0, 0)	8.027 [0.018]**	16.141 [0.006]***	20.636 [0.024]**							
3. r_ATB_60	0.077 (0.005)	14.18 [0.000]***	3.439 (0.371)	9.265 [0.000]***	(1, 0, 0)	8.424 [0.015]**	16.135 [0.006]***	20.321 [0.026]**							
4. r_ATB_120	0.077 (0.005)	14.085 [0.000]***	3.428 (0.370)	9.271 [0.000]***	(1, 0, 0)	8.101 [0.017]**	15.975 [0.007]***	20.255 [0.027]**							
5. r_ATB_180	0.077 (0.006)	13.971 [0.000]***	3.399 (0.368)	9.233 [0.000]***	(1, 0, 0)	7.909 [0.019]**	15.537 [0.008]***	19.899 [0.030]**							

The Table displays the lags of the independent variable in the model as $Var_0, 1, 2, 3$. The values are rounded up to the third decimal. Robust standard errors are provided in parentheses (). probability values are presented in square brackets [], and asterisks are used to indicate significance levels: * indicates $p < 0.10$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$

Table 5 GARCH Analysis: r_Price with ATB Variants

Model number / Exogenous variable	mu		ar (1)		ma (1)		ma (2)		Var_0		Var_1		Var_2		Var_3							
	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value						
1. r_Price	0.002 (0.002)	1.126 [0.260]			-0.067 (0.029)	-2.290 [0.022]**	0.036 (0.025)	1.456 [0.145]	0.097 (0.028)	3.48 [0.001]***					0.024 (0.023)	1.065 [0.287]						
2. r_ATB_30	0.002 (0.002)	1.453 [0.146]	-0.065 (0.028)	-2.299 [0.022]**					0.141 (0.051)	2.75 [0.006]***					0.045 (0.045)	1.001 [0.317]						
3. r_ATB_60	0.002 (0.002)	1.52 [0.128]	-0.064 (0.028)	-2.264 [0.024]**					0.215 (0.063)	3.416 [0.001]***			-0.047 (0.034)	-1.396 [0.163]								
4. r_ATB_120	0.002 (0.002)	1.548 [0.121]	-0.067 (0.029)	-2.349 [0.019]**					0.276 (0.058)	4.799 [0.001]***			-0.064 (0.030)	-2.092 [0.036]**								
5. r_ATB_180	0.002 (0.002)	1.620 [0.105]	-0.069 (0.029)	-2.426 [0.015]**																		
Model number / Exogenous variable	Omega		Alpha1		Beta1		Shape		Weighted Ljung-Box test—Standardized Residuals		Lag I		Lag II		Lag III		Lag 3		Lag 5		Lag 7	
	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value	Estimate	t value
1. r_Price	0.001 (0.001)	2.627 [0.009]***	0.081 (0.015)	5.560 [0.000]***	0.904 (0.014)	64.608 [0.000]***	4.546 (0.588)	7.730 [0.000]***	1.171 ^b [0.279]	4.030 ^{b,c} [0.062]*	5.794 ^{b,d} [0.292]	0.262 [0.608]	0.325 [0.933]	0.470 [0.981]								
2. r_ATB_30	0.001 (0.001)	2.689 [0.007]***	0.082 (0.016)	5.261 [0.000]***	0.903 (0.015)	61.009 [0.000]***	4.573 (0.621)	7.362 [0.000]***	1.343 ^a [0.247]	1.979 ^{a,e} [0.220]	5.141 ^{a,f} [0.095]*	0.253 [0.615]	0.306 [0.939]	0.446 [0.983]								
3. r_ATB_60	0.001 (0.001)	2.755 [0.006]***	0.081 (0.015)	5.360 [0.000]***	0.904 (0.015)	62.408 [0.000]***	4.681 (0.657)	7.127 [0.000]***	1.235 ^a [0.267]	1.988 ^{a,e} [0.217]	5.011 ^{a,f} [0.105]	0.259 [0.611]	0.296 [0.941]	0.438 [0.984]								
4. r_ATB_120	0.001 (0.001)	2.697 [0.007]***	0.080 (0.015)	5.310 [0.000]***	0.905 (0.014)	63.449 [0.000]***	4.619 (0.628)	7.356 [0.000]***	1.370 ^a [0.242]	2.296 ^{a,e} [0.131]	5.365 ^{a,f} [0.079]*	0.284 [0.594]	0.321 [0.935]	0.451 [0.983]								
5. r_ATB_180	0.001 (0.001)	2.641 [0.008]***	0.081 (0.015)	5.340 [0.000]***	0.905 (0.014)	63.833 [0.000]***	4.561 (0.612)	7.450 [0.000]***	1.397 ^a [0.237]	2.345 ^{a,e} [0.12]	5.419 ^{a,f} [0.076]*	0.288 [0.592]	0.324 [0.934]	0.448 [0.983]								

The Table displays the lags of the independent variable in the model as $Var_0[0,1,2,3]$. The values are rounded up to the third decimal. Robust standard errors are provided in parentheses (), probability values are presented in square brackets [], and asterisks are used to indicate significance levels: * indicates $p < 0.10$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$

^a 1 degree of freedom

^b 2 degrees of freedom

^c Lag $[2*(p+q) + (p+q)-1][5]$

^d Lag $[4*(p+q) + (p+q)-1][9]$, ^e Lag $[2*(p+q) + (p+q)-1][2]$, ^f Lag $[4*(p+q) + (p+q)-1][5]$

GARCH and forecasting results

This study used the optimal ARIMA models for the conditional mean of the GARCH processes to assess the GARCH(1,1) models (presented in Table 5). The results reveal that the exogenous variables in the conditional mean, AR (1), and MA(1) coefficients are significant. Moreover, all models demonstrated significant omega, alpha, and beta coefficients. The alpha coefficients range from 0.08 to 0.082, indicating a moderate level of persistence. This result suggests that recent volatility shocks have a notable but relatively modest impact on the current volatility. The beta coefficients range from 0.903 to 0.905, signifying that past volatility has a strong influence and persistence on current volatility. The values close to 1 also imply a clustering effect, where high or low volatility persists in subsequent periods. The significant omega coefficient in all cases indicates a baseline level of volatility, suggesting an inherent level of variance that other factors in the model cannot explain. Finally, the combined values of the alpha and beta coefficients do not exceed 1, satisfying the restriction of covariance stationarity. This result supports the stability of the models and suggests that the volatility process remains within a specific range over time. The examined models have robust results, as no autocorrelation and ARCH effects occur, as indicated by the weighted Ljung–Box tests on the standardized residuals and the ARCH-LM tests. Appendix 2 provides additional diagnostic tests for the examined models.

The additional forecasting performance metrics in Table 6 reveal that models 2, 3, 4, and 5 consistently exhibit lower error metrics than the benchmark model 1. Furthermore, the MCS test indicates that the benchmark model is excluded from the SSM for forecast horizons of 15, 12, 11, and 10 days.

Backtesting results

Entities “A” and “B” allocated identical amounts in USD to employ the DCA strategy for LINK trading. In contrast, entity “BH” initiated an investment equivalent to the cumulative investments that “A” and “B” made over the entire testing period, commencing on the initial day of the test period. Figure 5 displays the total USD amounts required for each trading frequency.

Note. For Entity “B,” this figure represents only the new funds injected, without considering any cash generated from token sales in previous periods. Entity “BH” invested the same amount as “A” and “B” but allocated the entire sum at the beginning of the testing period.

Figure 6 shows that the portfolios of entities “A” and “BH” consistently generated profits across all scenarios, irrespective of the invested DCA amount. These results serve as the benchmark for evaluating the effectiveness of entity B’s trading strategy.

In its trading strategy, entity “B” utilized the “events” defined in the methodology section as sell signals. Appendix 2 presents the differences and the second-order differences between the LINK price and ATP. Figure 7 depicts the graph displaying the total count of “events” per window size, reflecting the number of times entity “B” executed token sales.

Figure 7 illustrates the initial range of “events” across all four variables, starting from around 180 to 200 per variable for a window size of 3. These “events” gradually decreased at a similar pace until window 21; however, a sudden surge in the number of “events”

Table 6 Performance Metrics for ARIMA-GARCH Forecasting

Model number and exogenous variable	Performance metrics									
	Forecast horizon: 15					Forecast horizon: 12				
	MAE	MSE	RMSE	MAPE	MCS	AE	SE	MAE	MSE	RMSE
1. (r_Price)	0.0468	0.0031	0.0557	1.0143	0.7288			0.0542	0.0038	0.0615
2. r_ATB_30	0.0466	0.0031	0.0554	1.0131	1.0000	0.1230		0.0540	0.0037	0.0612
3. r_ATB_60	0.0466	0.0031	0.0554	1.0140	1.0000	1.0000		0.0540	0.0037	0.0611
4. r_ATB_120	0.0466	0.0031	0.0554	1.0143	1.0000	1.0000		0.0539	0.0037	0.0611
5. r_ATB_180	0.0466	0.0031	0.0554	1.0144	1.0000	1.0000		0.0540	0.0037	0.0611
Performance metrics										
Model number and exogenous variable	Forecast horizon: 10					Forecast horizon: 9				
	MAE	MSE	RMSE	MAPE	MCS	AE	SE	MAE	MSE	RMSE
	MAE	MSE	RMSE	MAPE	MCS	AE	SE	MAE	MSE	RMSE
1. (r_Price)	0.0527	0.0034	0.0585	1.0188	0.5664			0.0480	0.0028	0.0530
2. r_ATB_30	0.0525	0.0034	0.0583	1.0162	1.0000			0.0478	0.0028	0.0528
3. r_ATB_60	0.0524	0.0034	0.0582	1.0172	1.0000	0.1824		0.0478	0.0028	0.0527
4. r_ATB_120	0.0524	0.0034	0.0582	1.0176	1.0000	1.0000		0.0478	0.0028	0.0527
5. r_ATB_180	0.0524	0.0034	0.0582	1.0177	1.0000	1.0000		0.0478	0.0028	0.0527

Table 6 (continued)

Model number and exogenous variable	Performance metrics											
	Forecast horizon: 7						Forecast horizon: 6					
	MAE	MSE	RMSE	MAPE	MCS	SE	MAE	MSE	RMSE	MAPE	MCS	SE
1. (r_Price)	0.0506	0.0032	0.0564	1.0288	0.5878	0.5030	0.0460	0.0027	0.0518	1.0371	1.0000	0.6766
2. r_ATB_30	0.0504	0.0031	0.0561	1.0257	1.0000	1.0000	0.0458	0.0027	0.0516	1.0347	1.0000	1.0000
3. r_ATB_60	0.0503	0.0031	0.0560	1.0274	1.0000	1.0000	0.0458	0.0027	0.0515	1.0370	1.0000	1.0000
4. r_ATB_120	0.0503	0.0031	0.0560	1.0280	1.0000	1.0000	0.0458	0.0027	0.0515	1.0378	1.0000	1.0000
5. r_ATB_180	0.0504	0.0031	0.0560	1.0282	1.0000	1.0000	0.0458	0.0027	0.0515	1.0381	1.0000	1.0000

The p -values for the MCS tests are presented at a significance level of $\alpha = 10\%$. The MCS tests assess Absolute Errors (AE) and Squared Errors (SE) loss functions. The MCS test results are reported for all forecast horizons except when the test is inapplicable for a horizon of 1 because of limitations due to the test parameters for forecast horizons less than six. An absent p -value in the MCS results signifies the exclusion of the model from the Superior Set of Models.

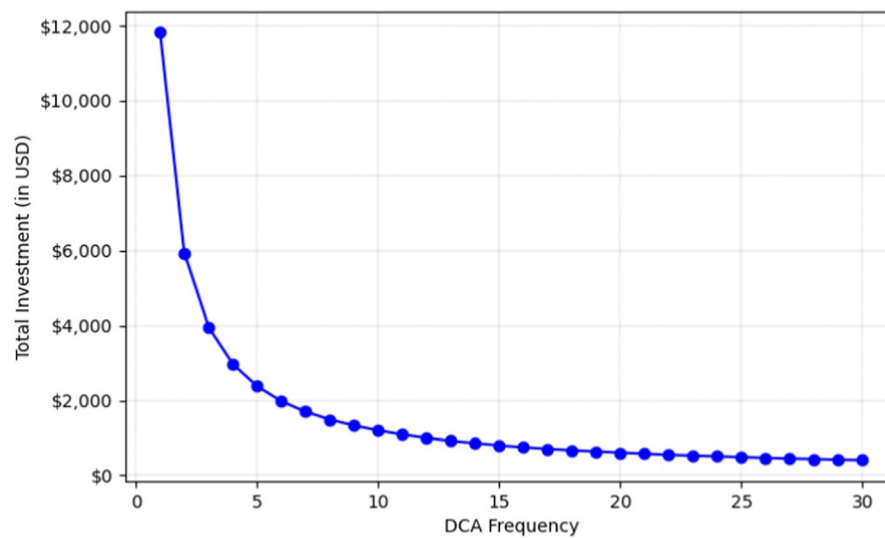


Fig. 5 Aggregate Investment Amount by DCA Frequency for Entities “BH,” “A,” and “B”

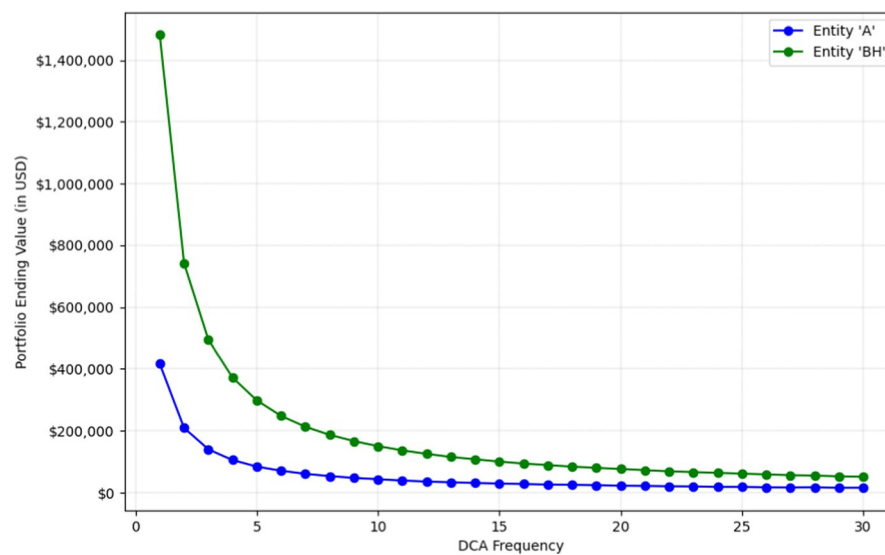


Fig. 6 Entity “BH” and “A” portfolio ending value. *Note.* For entity “BH,” the graph displays the portfolio’s final value, achieved by investing the total amount that entity “A” invested using the DCA method, starting on the first day of the test period

occurred at window size 22, ranging from approximately 60 to 70. Following this spike, the rate at which the “events” occur consistently declined.

Regarding the trading strategy results, entity “B” demonstrated better performance than control entity “A” in 564 cases; it also outperformed entity “BH” in 7 instances (Appendix 2). Figures 8, 9, 10, and 11 display the outcomes of the trading strategies employed by entity “B” versus entity “A.”

Figure 8 illustrates the results, indicating that for a DCA frequency of “1,” positive outcomes were observed in 24 out of 28 window sizes. Similarly, with a DCA frequency of “19,” positive results were achieved in 22 out of 28 window sizes. The results vary between 0 and 10 in window sizes for other DCA frequencies. Notably, when

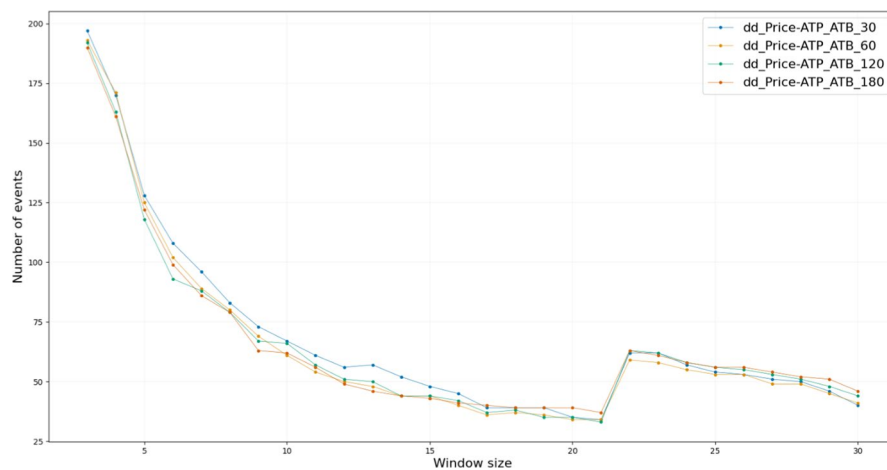


Fig. 7 Event frequency based on Window Size and Variable

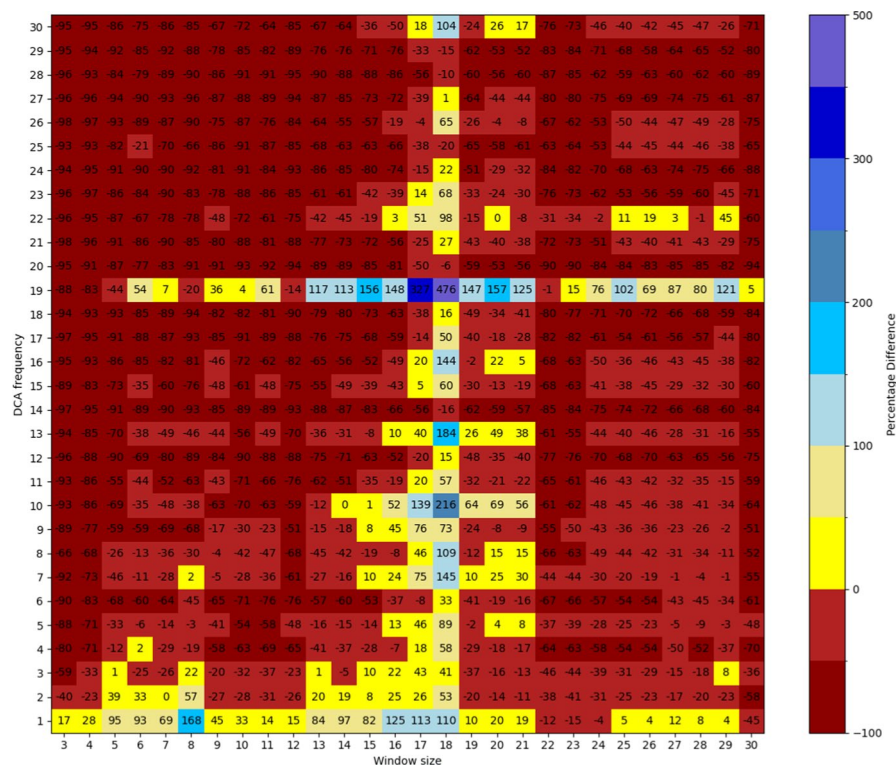


Fig. 8 Percentage Difference in Results: Entity "A" vs. Entity "B" with dd_Price-ATP 30

considering a window size of "18," significant results are evident in 25 out of 30 DCA frequencies. This result suggests a weak clustering of favorable outcomes within the range of window sizes from 13 to 21.

Figure 9 shows the positive results for a DCA frequency of "1" in 22 out of the 28 window sizes. Similarly, 23 out of 28 window sizes exhibited positive outcomes for a DCA frequency of "19." Additionally, scattered positive results were found for various DCA

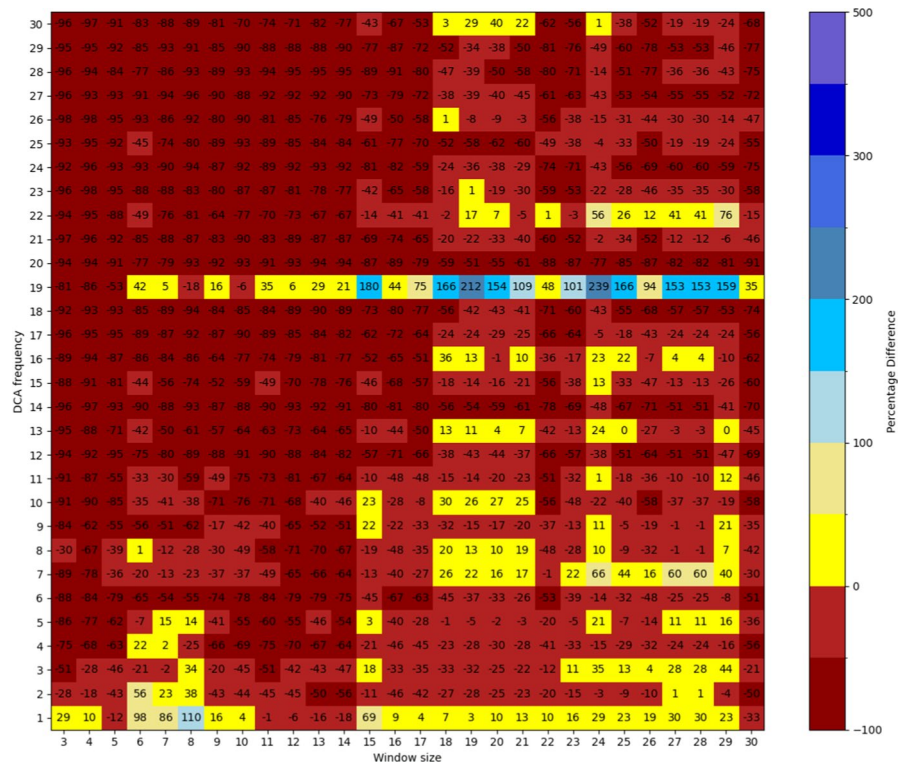


Fig. 9 Percentage Difference in Results: Entity "A" vs. Entity "B" with dd_Price-ATP 60

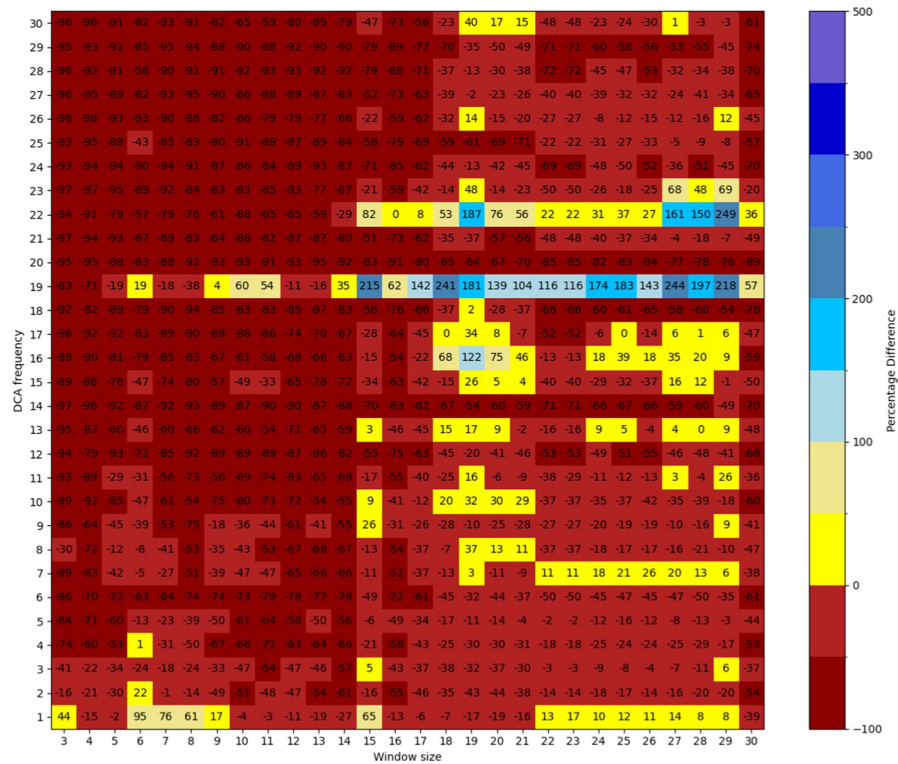


Fig. 10 Percentage Difference in Results: Entity "A" vs. Entity "B" with dd_Price-ATP 120

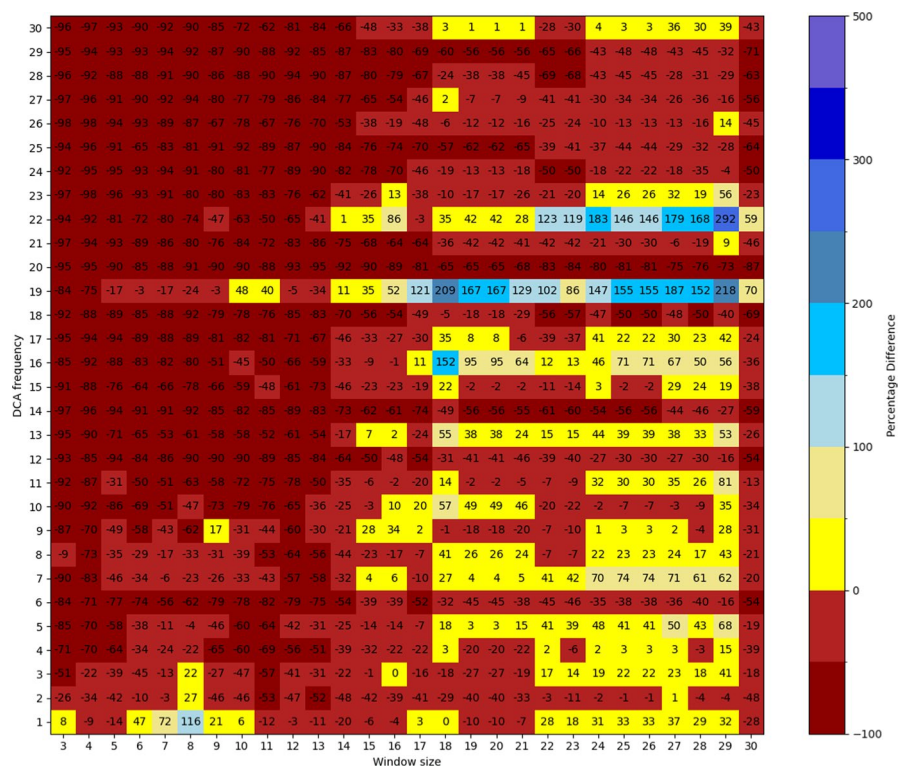


Fig. 11 Percentage Difference in Results: Entity "A" vs. Entity "B" with dd_Price-ATP 180

frequencies ranging from "2" to "22," particularly for window sizes between "15" and "29."

Figure 10 presents the results, indicating positive outcomes for a DCA frequency of "1" in 14 out of 28 window sizes. Furthermore, when considering a DCA frequency of "19," positive results are found in 22 out of 29 window sizes. Additionally, within the window size range of 15 to 29, scattered positive results for DCA frequencies range "7–17" and "22."

Figure 11 illustrates positive results in 16 out of 28 window sizes for a DCA frequency of "1." Similarly, for a DCA frequency of "19," positive outcomes occurred in 19 out of 28 window sizes. Notably, the DCA frequency "22" shows positive results in 12 out of 28 window sizes. Furthermore, a noticeable clustering of positive results exists for window sizes ranging from 15 to 29 and for DCA frequencies between "3" and "17."

Figures 12, 13, 14 and 15 present the trading strategy results for entity "BH" versus entity "B."

Figure 12 illustrates two positive results in the DCA frequency "19" and the window sizes "17" and "18," for +29% and +73%, respectively.

Figure 13 shows one positive result of +2% in the DCA frequency "19" and window size "24."

Figure 14 shows 3 positive results ranging from +3% to +5% for the DCA frequencies "19" and "22."

Figure 15 shows one positive result of +17% for the DCA frequency "22" and window size "29."

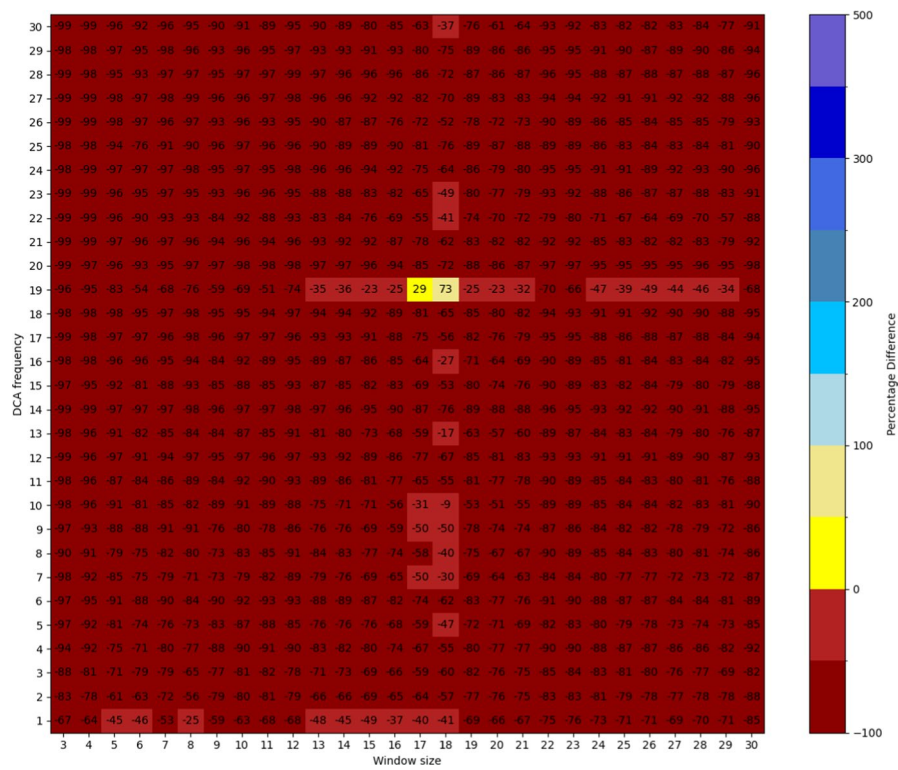


Fig. 12 Percentage Difference in Results: Entity "BH" vs. Entity "B" with dd_Price-ATP 30

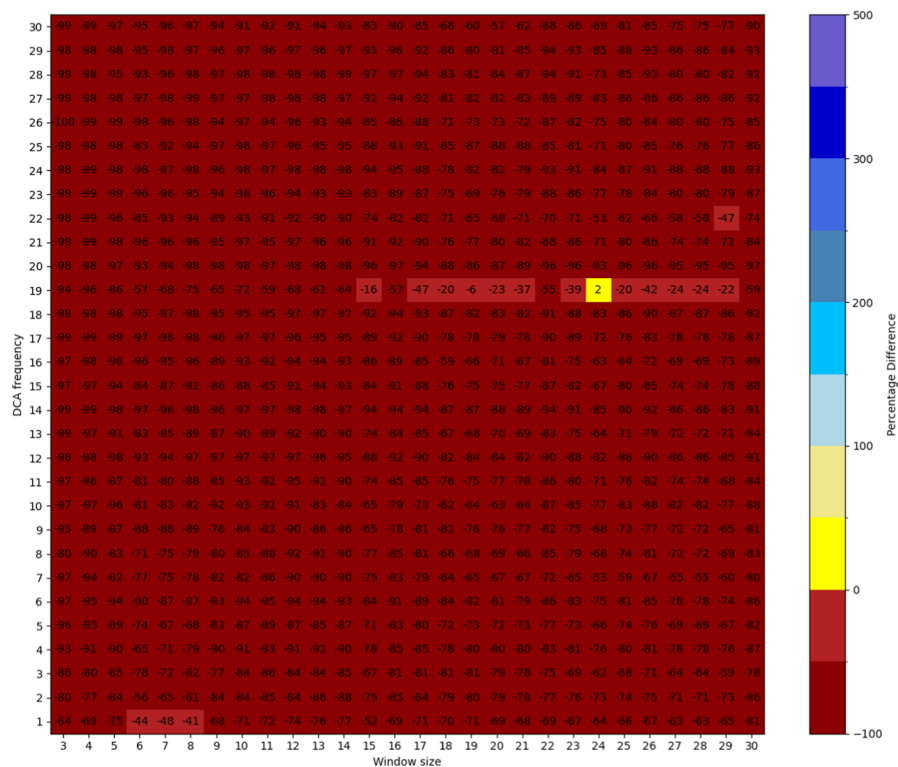


Fig. 13 Percentage Difference in Results: Entity "BH" vs. Entity "B" with dd_Price-ATP 60

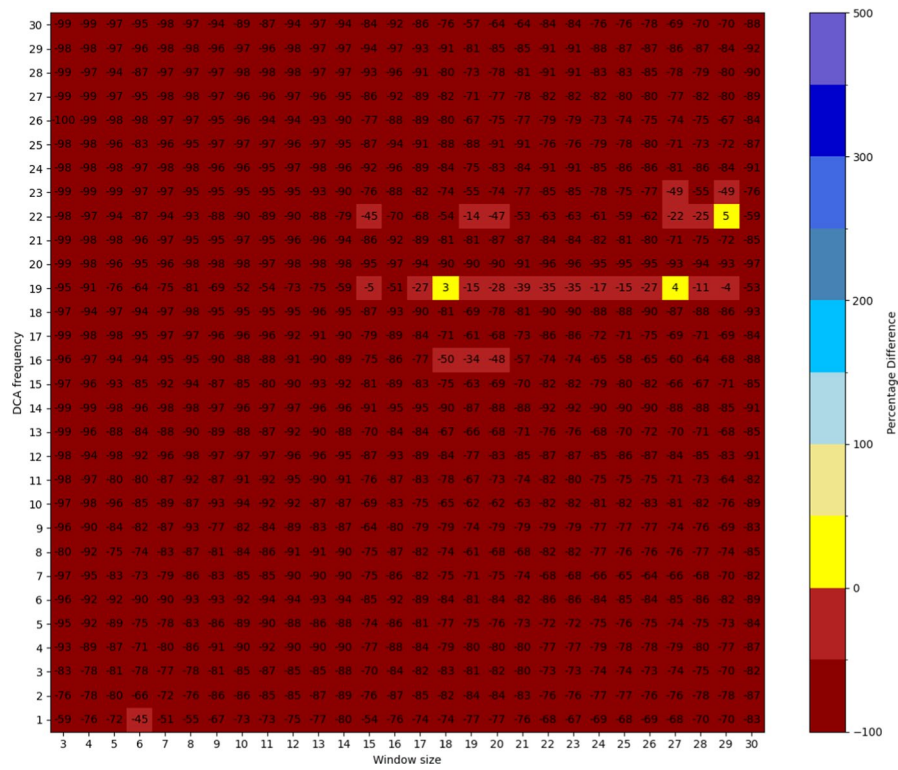


Fig. 14 Percentage Difference in Results: Entity "BH" vs. Entity "B" with dd_Price-ATP 120

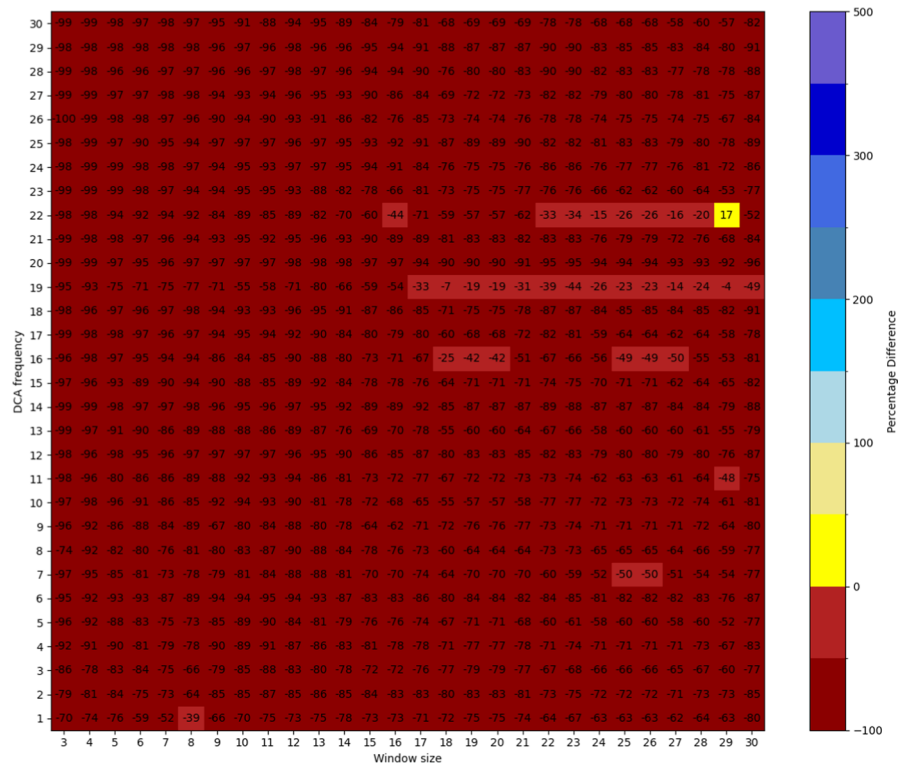
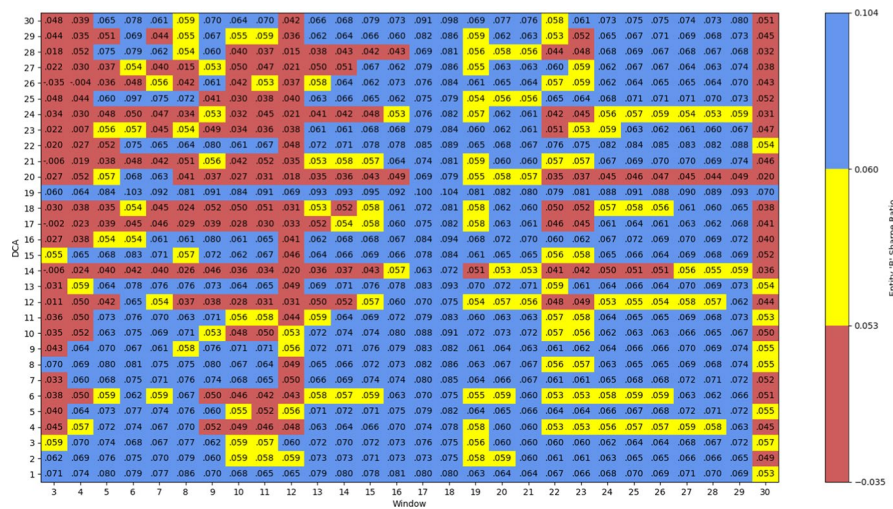
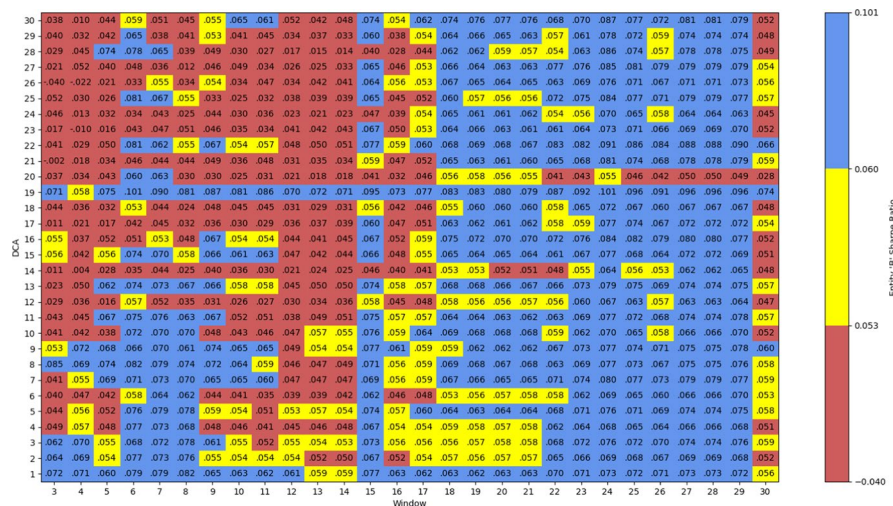


Fig. 15 Percentage Difference in Results: Entity "BH" vs. Entity "B" with dd_Price-ATP 180

Table 7 Sharpe Ratios for Strategies of Entities “BH” and “A”

Strategy	Sharpe ratio	Strategy	Sharpe ratio	Strategy	Sharpe ratio	Strategy	Sharpe ratio
Buy-and-hold	0.053	DCA 8	0.060	DCA 16	0.060	DCA 24	0.060
DCA 1	0.060	DCA 9	0.060	DCA 17	0.060	DCA 25	0.060
DCA 2	0.060	DCA 10	0.060	DCA 18	0.060	DCA 26	0.060
DCA 3	0.060	DCA 11	0.060	DCA 19	0.060	DCA 27	0.060
DCA 4	0.060	DCA 12	0.060	DCA 20	0.060	DCA 28	0.060
DCA 5	0.060	DCA 13	0.060	DCA 21	0.060	DCA 29	0.060
DCA 6	0.060	DCA 14	0.060	DCA 22	0.060	DCA 30	0.060
DCA 7	0.060	DCA 15	0.060	DCA 23	0.060		

**Fig. 16** Sharpe ratios: Entity “B” with dd_Price-ATP 30**Fig. 17** Sharpe ratios: Entity “B” with dd_Price-ATP 60

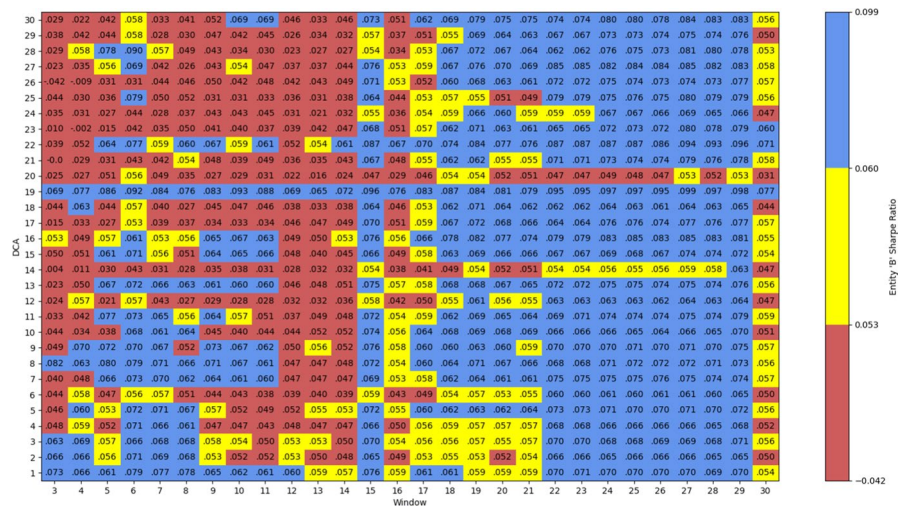


Fig. 18 Sharpe ratios: Entity "B" with dd_Price-ATP 120

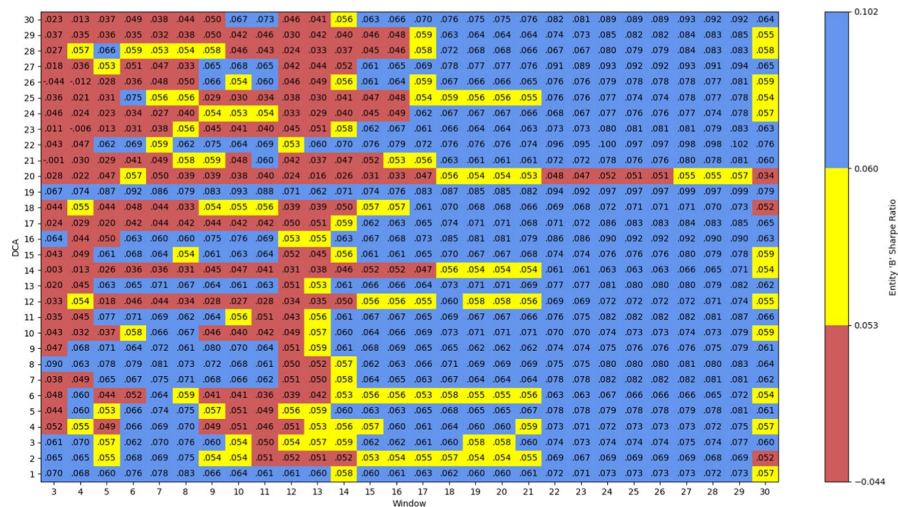


Fig. 19 Sharpe ratios: Entity "B" with dd_Price-ATP 180

Table 7 continues the comparison among the examined entities, presenting the daily Sharpe ratios for the strategies of "BH" and "A." Notably, the values are rounded to three decimal places, with entity "A" having a ratio of 0.60 and entity "BH" having a ratio of 0.53; therefore, entity "A" achieves relatively better Sharpe ratios than entity "BH." The comparison proceeds by examining the Sharpe ratios of each computed trading strategy utilizing the ATP variations, as shown in Figs. 16, 17, 18, and 19.

Figure 16 shows that the Sharpe ratios for the trading strategy of entity "B" using "dd_Price-ATP 30" are higher than entity "A" in 474 cases and higher than entity "BH" in 616 cases.

Figure 17 shows that the Sharpe ratios for the trading strategy of entity "B" using "dd_Price-ATP 60" are higher than entity "A" in 412 cases and higher than entity "BH" in 556 cases. Figure 18 reveals that the Sharpe ratios for the trading strategy of entity "B" using

“dd_Price-ATP 120” are higher than entity “A” in 420 cases and higher than entity “BH” in 551 cases.

Figure 19 shows that the Sharpe ratios for the trading strategy of entity “B” using “dd_Price-ATP 180” are higher than entity “A” in 493 cases and higher than entity “BH” in 619 cases.

The Sharpe ratio results indicate that the proposed strategies perform much better than the benchmark models. These results suggest the potential for enhanced portfolio returns, indicating that the developed strategies could offer a more efficient allocation of capital by achieving higher returns per unit of risk.

Discussion

This paper’s ATB and AMC methodologies show a more modest approach to crypto-asset valuation. The effect of token dormancy is measured and expressed regarding the valuation of LINK, which gives more insight into the token’s activity. Alternative methods like ASOL represent the average lifespan of spent outputs and do not consider the unspent outputs that may be inside the same wallet. Additional methods include CDD, which measures the number of days since the creation of spent outputs, or HODL waves that divide the dormant Bitcoins into age bands. In contrast to these approaches, the ATB method observes the daily inactivity of all the tokens in account-based models. Additionally, the ATB method summarizes the tokens’ activity in a single metric, providing a customizable formula to calculate the overall token dormancy at any specific time. Furthermore, this dormancy metric can calculate the value of a CA token and establish the ATP fundamental price, which can create a trading “sell” signal using its difference to the market price.

Nonetheless, there are some limitations when implementing this methodology. The current method may be unsuitable for stablecoins, which may have unique dynamics and assumptions, or privacy coins, like Monero, where retrieving the balances of the addresses in a private network is currently impossible. Another limitation of this study’s methodology is that some entities could still be interested in their tokens and could have staked them for additional interest; however, limitations in time-locked smart contracts make it impossible to move these tokens, rendering them dormant and excluding them from the AMC. Furthermore, some entities might be interested in a token they own in their addresses and would still interact with the smart contract that issued it; however, such activity would not be captured in the transactions proposed in this research paper. Therefore, the modified ATB and AMC versions could capture these effects.

The ARIMA-GARCH models showed that using the ATB as exogenous variables could be helpful when capturing underlying dynamics, enriching the model’s predictive capabilities for the LINK returns—at least for the tested UL values in this manuscript. The forecasting results showed marginally lower values of performance metrics in most cases compared to the benchmark model. The results of the MCS test showed a marginally better performance for the examined models than for the benchmark model. This finding could indicate that the ATB variables could help provide better price return forecasts for the examined period; however, further analysis in multiple periods could support these results.

The backtesting analysis revealed that ATP exhibits potential utility when providing a selling indicator, as entity “B” had a good performance compared to entity “A” in many cases but underperformed against entity “BH.” The assessment was primarily conducted using the four different UL values and a DCA buying signal; this approach could be integrated with other indicators in future applications, warranting further investigation and experimentation. The Sharpe ratios for the backtested strategies are often higher than the benchmark strategies, indicating a better risk-adjusted return. Finally, the backtesting results suggest that the ATB time series could be valuable when incorporated as an indicator in a customized strategy.

Nonetheless, this study’s backtesting methodologies are a theoretical approach for assessing diverse trading strategies for crypto-assets—their scope is limited by the absence of a buying signal counterpart to a selling event signal. First, the DCA method was applied with fixed buying intervals; incorporating other factors might generate more optimal “buy” signals. Second, a buy-and-hold strategy comparison may not always be appropriate for crypto-assets. The frequent occurrence of exploding time series and volatility clustering or scenarios where airdrops, early mining, or buying opportunities allow minimal initial investment in a crypto-asset could introduce bias in selecting the starting point. Third, some crypto-assets offer staking, which can yield additional token returns, representing an opportunity cost that should be considered in the overall strategy; however, the token examined in this study did not have a staking reward ability during the examined period. Finally, varying prices across multiple exchanges operating a 24-h, different commission rates for various investor categories, and distinct tax laws and jurisdictions can result in significant variations in the final buying cost of a CA and, thus, the subsequent profits or losses.

Including active supply as a fundamental metric holds significant potential for various aspects of cryptocurrency analysis. First, this study shows that active supply can reflect investor behavior. Periods of lower active supply may indicate investors’ tendency to hold the asset, reducing price volatility. Conversely, growth in the active token supply could signal investors’ willingness to sell, creating selling pressure and increasing market supply. This increased supply liquidity can result in heightened price volatility due to downward selling pressures, especially if the market lacks sufficient depth. Additionally, while price and volume are critical indicators, the dynamics of active supply offer valuable insights into the network health of a cryptocurrency. These dynamics reflect the assets’ utility and adoption, as well as the profiles of their holders, distinguishing short-term traders from long-term investors.

Conclusion

This research paper presented the ATB methodology, the proposed AMC valuation method, the ATP fundamental value, an ARIMA-GARCH analysis, and a trading strategy. This research provides an alternative to the current valuation methods and considers the activity of tokens, proposing a new method to calculate the dormancy of a network’s tokens. Furthermore, the ATB variables were statistically significant in the examined models, providing marginally better forecasting results than the benchmark model. Finally, the backtested trading strategy was proposed using the ATP price, indicating a real-world application of the new fundamental value.

This study's methodology can be applied to tokens other than LINK in the Ethereum blockchain by calculating the ATB values using the balances and dormancy of the examined token in the wallet addresses that contain it.

Improved versions, including more interactions with smart contracts, could also provide a more comprehensive representation of the token holder activity effects in a token's native or cross-chain smart contract environment, considering the token's holder activity in multiple blockchains. Modified versions of this methodology could be proposed for other crypto-assets, like nonfungible tokens (NFTs), considering NFTs' unique characteristics, exchanges, and prices in their respective collections. The activity metric for NFTs could reflect market demand, community engagement, price speculation, and an artist's popularity. Another extension of this method could be analyzing governance smart contracts to measure a community's participation. The active tokens of the involved wallets in the decision-making process can be calculated, indicating the project's stability, users' confidence in it, overall governance satisfaction, and the impact of the governance proposal, where an important decision could relate to more activity and vice versa. Applying this study's methodology to other tokens could emphasize the significance of token activity as a valuation tool, examining its impact under different market conditions and periods and its relationship with market cycles. High and steady active token balance levels might suggest a secure environment, indicating user confidence in the token or the underlying protocol. Furthermore, future studies can analyze the activity metric concerning token staking and how their interactions affect its price. Finally, the relationship between token activity, liquidity, and risk could be examined as indicators of token price manipulation and entry and exit points.

This study provides new insights into evaluating a crypto-asset and could improve understanding of the connection between on-chain data and the value of a token.

Appendix 1: Formulas

Dormancy penalty: Normalization approach

One may use a unity-based normalization approach to calculate the dormancy penalty, as in Formula (22).

$$P_{i,t,UL} = \frac{\min(Dormancy_{i,t}, Useful\ life_{CA}) - 0}{Useful\ life_{CA} - 0} \Leftrightarrow \frac{\min(Dormancy_{i,t}, Useful\ life_{CA})}{Useful\ life_{CA}} \quad (22)$$

In Formula 22, the dormancy value may have a higher value than the UL; however, to ensure that the score lies within the range from zero to one, the minimum between the dormancy and the UL values should be used as the value in the equation to be normalized.

Performance metrics

The forecasting results are evaluated using the following performance metrics formulas: (23) for MAE, (24) for MSE, (25) for RMSE, and (26) for MAPE.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (23)$$

$$MSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (24)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (25)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (26)$$

where “ y ” is the actual value and “ $\hat{y}$ ” is the estimated value for “ i ” observations. The MAE evaluates discrepancies between actual and predicted values by considering their absolute difference. The MSE assigns greater penalties to larger errors than the MAE; however, it may be more sensitive to outlier influence due to squaring these differences. The RMSE, derived from the square root of MSE, offers another perspective and is a standard performance metric in financial literature. The MAPE presents errors in percentage terms, facilitating interpretation and accommodating both positive and negative deviations. In all instances, lower metric values closer to zero across all measures signify higher prediction accuracy.

Appendix 2: Supplementary Tables and Figures

Table 8 presents additional diagnostics for the estimated models in this study.

The diagnostic tests conducted on the estimated models reveal robust results for the examined models. The Weighted Ljung-Box Test on Standardized Squared Residuals indicates no significant autocorrelation, suggesting the model effectively captures the variance structure. The absence of null rejection for the Adjusted Pearson Goodness-of-Fit Test confirms a good fit for the t-Student distribution selection for the models. The Nyblom Stability Test demonstrates parameter stability over time, indicating model consistency, as the joint statistic values did not exceed the asymptotic joint statistic values. Finally, the non-rejection of the Sign Bias Test null hypothesis indicates that the models adequately explain large positive or negative shocks.

Figure 20 demonstrates the difference between ATP values and the market price of LINK. These observations suggest that the overall token activity is more pronounced during the initial stages of its existence.

Figure 21 displays the plots illustrating the differences of differences between the market price of LINK and ATP variants. Initially, these differences exhibit small values, which is expected due to the relatively low price of LINK. However, as time progresses, these values increase, indicating a larger magnitude of differences. Notably, there is evidence of volatility clustering towards the end of the examined period. The difference values in Fig. 21. serve as input data in Formula (20).

Table 8 Additional diagnostics

Model number / Exogenous variable	Test							
	Weighted Ljung-Box Test on Standardized Squared Residuals			Adjusted Pearson Goodness-of-Fit Test				
	Lag			Group				
	1	5	9	20	30	40	50	
1. (r_Price)	0.00 [0.99]	0.54 [0.95]	0.78 [0.99]	25.75 [0.14]	33.28 [0.27]	36.27 [0.59]	43.30 [0.70]	
2. r_ATB_30	0.01 [0.94]	0.47 [0.96]	0.70 [1.00]	22.71 [0.25]	34.41 [0.22]	41.80 [0.35]	43.18 [0.71]	
3. r_ATB_60	0.01 [0.94]	0.43 [0.97]	0.65 [1.00]	27.80 [0.09]*	28.86 [0.47]	51.84 [0.08]*	58.02 [0.18]	
4. r_ATB_120	0.01 [0.94]	0.44 [0.97]	0.65 [1.00]	21.22 [0.32]	25.75 [0.64]	27.12 [0.92]	41.86 [0.76]	
5. r_ATB_180	0.00 [0.95]	0.44 [0.97]	0.65 [1.00]	19.29 [0.44]	27.88 [0.52]	28.88 [0.88]	36.95 [0.90]	
Model number / Exogenous variable	Nyblom stability test				Sign Bias Test			
	Joint Statistic	Joint Statistic Asymptotic Critical Values			Sign Bias	Negative Sign Bias	Positive Sign Bias	Joint Effect
		10%	5%	1%				
1. (r_Price)	1.62	1.69	1.90	2.35	0.31 [0.76]	0.32 [0.75]	0.27 [0.79]	0.85 [0.84]
2. r_ATB_30	1.83	1.89	2.11	2.59	0.45 [0.65]	0.50 [0.62]	0.14 [0.89]	1.38 [0.71]
3. r_ATB_60	1.82	1.89	2.11	2.59	0.44 [0.66]	0.50 [0.62]	0.13 [0.90]	1.33 [0.72]
4. r_ATB_120	1.70	1.89	2.11	2.59	0.36 [0.72]	0.56 [0.58]	0.19 [0.85]	1.36 [0.72]
5. r_ATB_180	1.61	1.89	2.11	2.59	0.38 [0.70]	0.51 [0.61]	0.21 [0.84]	1.33 [0.72]

The asterisk indicates the significance level: * $p < 0.1$. The square brackets indicate the corresponding p -values, while those without brackets represent the statistics values. For the Weighted Ljung-Box Test on Standardized Squared Residuals tests, the degrees of freedom are equal to 2

Figures 22 and 23 are reported as three-dimensional plots, showing the strategies where Entity “B” brings better results compared to Entities “A” and BH” respectively.

Appendix 3: Supplementary Explanatory Notes

ERC-20 tokens

The “Ethereum request for comment” 20 (ERC-20) is a standard interface for tokens within smart contracts, enabling the implementation of a standardized application programming interface (API). This interface facilitates basic token functionalities such as token transfers and approvals for spending by on-chain third parties. In addition, this standard was created to promote the reusability of tokens on the Ethereum blockchain, allowing them to be utilized by various applications, ranging from wallets to decentralized exchanges (Vogelsteller and Buterin 2015). ERC-20 tokens were used extensively for

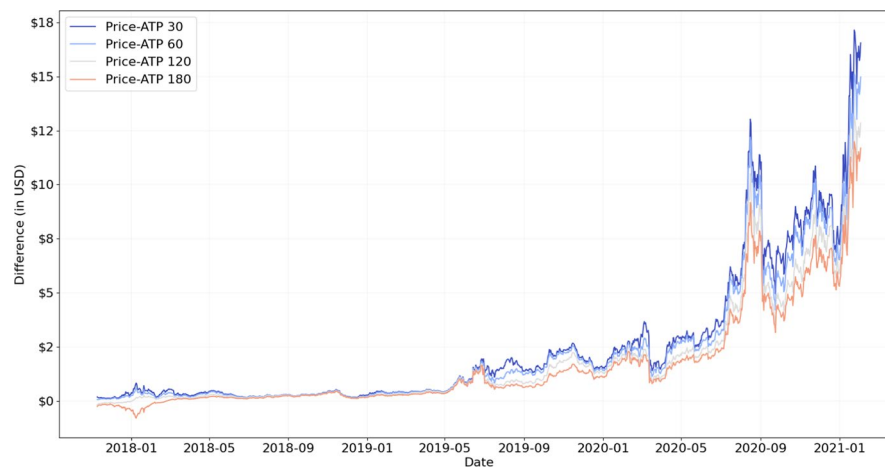


Fig. 20 Difference between LINK market price and ATP variants over time

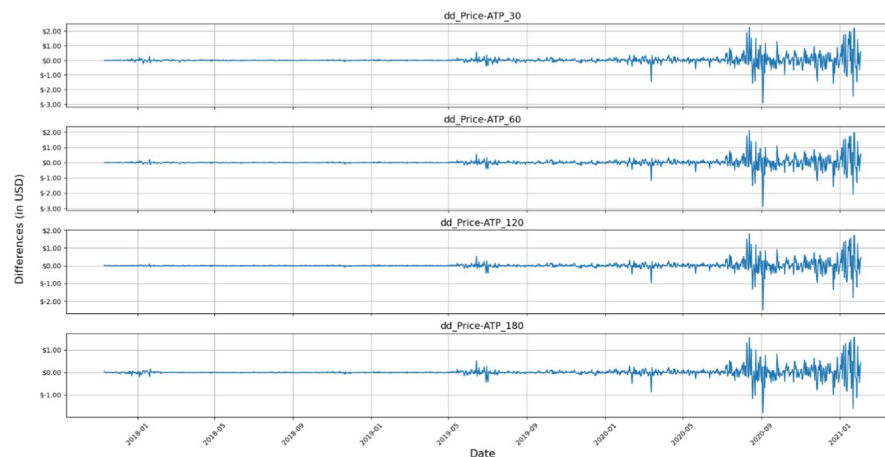


Fig. 21 Differences of differences between LINK Market Price and ATP variants over time

crowdfunding projects on the Ethereum blockchain in an unregulated manner in most countries (Victor and Lüders 2019). As of June 9th, 2023, the Ethereum network has witnessed the existence of over 650,000 ERC-20 tokens and a cumulative transaction count of around 1.25 billion (Blockchair.com, n.d.), which underscores the vital role played by the ERC-20 token standard within the Ethereum ecosystem.

Chainlink: a decentralized oracle network

Chainlink is a decentralized oracle network that provides data from the real world to decentralized smart contract applications deployed on various blockchains, like Ethereum (Breidenbach et al., n.d.). The Chainlink token functions as the native currency of the Chainlink network, which operates on the Ethereum blockchain, and it also serves as a payment method for data providers within the Chainlink network. As of March 2023, Chainlink stands out as the leading decentralized oracle network regarding market capitalization within the crypto-asset industry. Its primary role involves

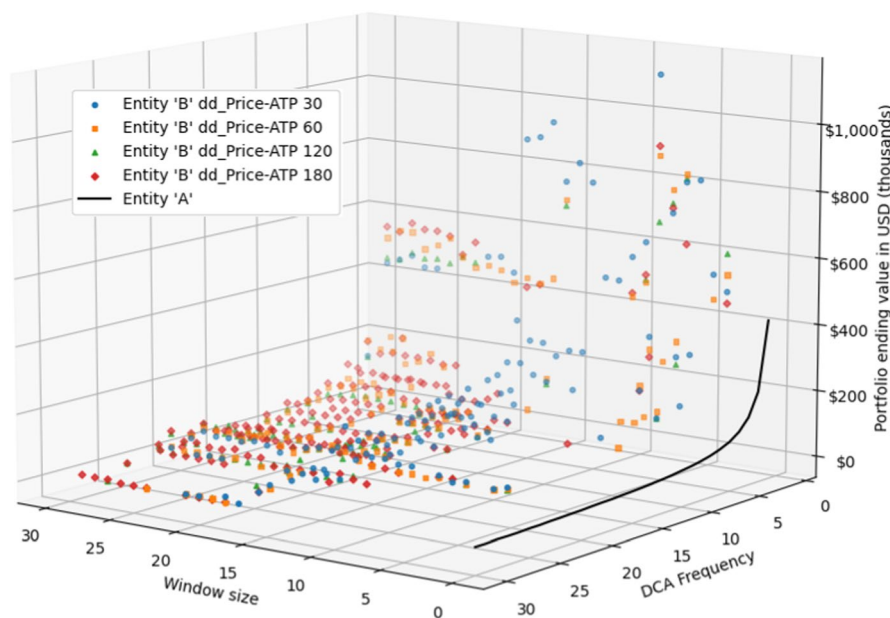


Fig. 22 Entity "A" vs. Entity "B" best strategies. *Note.* In this Figure, only the strategies affiliated with Entity "B" that exhibit superior total returns compared to those of Entity "A" are showcased

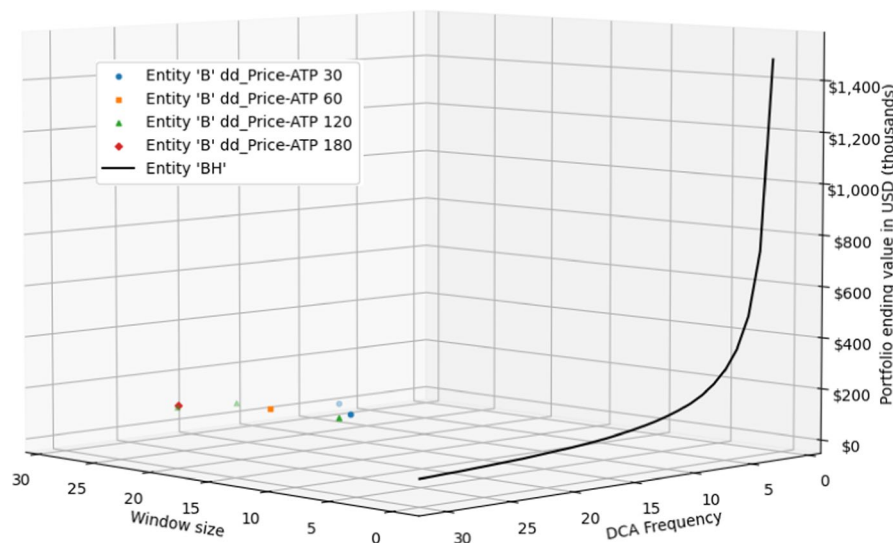


Fig. 23 Entity "BH" vs. Entity "B" best strategies. *Note.* This Figure presents only the optimal strategies where Entity "B" surpasses Entity "A" and Entity "BH."

supplying real-time data to smart contracts deployed on various blockchain networks, facilitating decentralized applications. Chainlink was initially launched on the Ethereum network on September 16th, 2017, according to Blockchair.com, and since then, it has been consistently expanding and evolving its operations. Therefore, the Chainlink token was selected to demonstrate the ATB method because I deemed it a sufficiently essential token for the reasons mentioned. The smart contract associated with the LINK token can be identified by its address "0×514910771AF9Ca656af840dff83E8264EcF986CA".

Abbreviations

AB	Activity-scored address Balance
ADF	Augmented Dickey-Fuller
AE	Absolute Error
AMC	Active Market Capitalization
API	Application Programming Interface
ARCH	Autoregressive Conditional Heteroskedasticity
ARIMA	Autoregressive Integrated Moving Average
AS	Activity Score
ATB	Active Token Balance
ATP	Active Token Price
BTC	Bitcoin
CA	Crypto-Asset
CS	Circulating Supply
CDD	Coin Days Destroyed
DCA	Dollar Cost Averaging
DT	Deposited Tokens
EPA	Equal Predictive Ability
ERC	Ethereum Request for Comment
FDV	Fully Diluted Valuation
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
JB	Jarque-Bera
KPSS	Kwiatkowski–Phillips–Schmidt–Shin
LINK	Chainlink (token)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percent Error
MC	Market Capitalization
MCS	Model Confidence Set
MP	Market Price
MSE	Mean Squared Error
NFT	Non-Fungible Token
PoS	Proof of Stake
PoW	Proof of Work
RMSE	Root Mean Squared Error
SE	Squared Error
SSM	Superior Set of Models
TrS	Transaction Status
TS	Total Supply
UL	Useful Life
USD	United States Dollar
UTXO	Unspent Transaction Output

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Declarations

Competing interests

The author declares no competing interests.

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